Exam 1

Your Name:

Instructions

Solve each of the following problems to the best of your abilities. The exam is worth 100 points and is calibrated for 90 minutes. You have the full two hours to work on the exam.

Once you have completed the exam, hand it to me, and you can take a break before the second part of class.

Good luck!

Problem 1

(25 points) An electric point charge with a strength of +1.5 nC is placed at the point (2.4 m, 3.3 m). A second electric point charge with a strength of –3.2 nC is placed at the point (-1.7 m, 5.5 m).

a) (5 points) What is the force that the +1.5 nC charge exerts on the –3.2 nC charge? Remember that force is a vector!
b) (5 points) What is the force that the –3.2 nC charge exerts on the +1.5 nC charge? Remember that force is a vector!
c) (5 points) What is the net electric field produced by the two point charges at the origin? Remember that electric field is a vector!
d) (5 points) Draw a sketch of the electric field lines around the two point charges. Be sure to place the charges accurately on a coordinate grid and specify the field strength with the relative density of field lines.
e) (5 points) Are these two point charges in space an example of an ideal electric dipole? Why or why not?

Problem 2

(30 points) A non-uniform electric field is given by $\vec{E} = 10x\ \hat{z}\ \frac{N}{C}$. A rectangular plate is placed in the xy-plane such that its left edge is along the y-axis, its bottom edge is along the x-axis, and its corner is at the origin. The plate has a length of 1.1 meters along the y-direction and 0.6 meters along the x-direction. Let's calculate the net electric flux passing through the plate.

a) (5 points) Start off by drawing a diagram of the setup described above. Be sure to label the coordinate axes, the plate, and the electric field. Additionally, label the area vector of the plate.
b) (5 points) We should cut up the plate into tiny rectangles, each with an area of dA. Sketch a rectangular area dA on your diagram, and write out an expression for each tiny area dA. Think carefully about how those rectangles should be oriented!
c) (5 points) Next, we should calculate the electric flux through each tiny area dA. Write out an expression for each tiny flux $d\phi$.
d) (5 points) We can sum up (i.e. integrate over) all of the tiny fluxes to get the total electric flux. Write out an integral for the total electric flux. Be sure to include the limits of integration.
e) (5 points) Solve your integral from part (d) to get the total electric flux passing through the plate.
f) (5 points) Suppose that the electric field in this region of space was pointing in the +y direction rather than the +z direction. How would this change your total calculated flux?

Problem 3

(25 points) A disk of charge with a charge density of $+\sigma$ and a radius of R is placed in the xy-plane with its center along the z-axis. The electric field produced by the disk along the positive z-axis is given by:

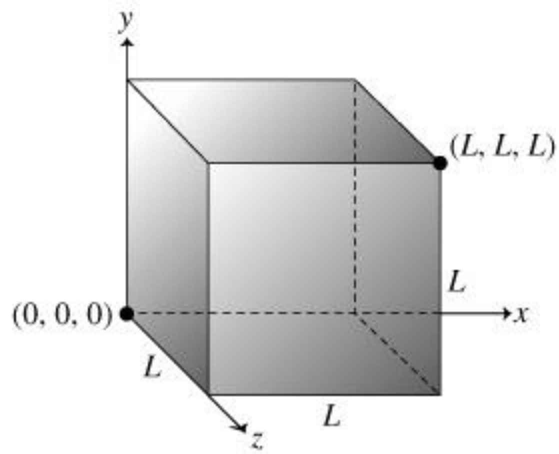
$$\vec{E} = \frac{\sigma}{2\epsilon_o}\left(1 - \frac{z}{\sqrt{R^2 + z^2}}\right)\hat{z}$$

a) (5 points) What is the electric field at the location z = R?
b) (5 points) Suppose I take the limit where the distance along the axis, z, is much larger than the radius, R, of the disk. What is the expression for the electric field?
c) (5 points) Suppose I take the limit where the radius, R, of the disk is much larger than the distance along the axis, z. What is the expression for the electric field?
d) (5 points) Recall that the kinematics equations that we learned in physics work for a particle undergoing constant acceleration. Explain why the kinematics equations are not valid for the motion of a point charge along the z-axis in this system. Be specific!
e) (5 points) Write an expression for the work done by the electric field as it moves a positively charged particle (+Q) from z = 1 to z = 2. You do NOT have to solve the integral. Simply write out the integral, simplify the dot product, and label the limits of integration.

$$W = \int_a^b \vec{F} \cdot d\vec{x}$$

Problem 4

(20 points) A uniform electric field is given by E = 100 N/C and points in the +z direction. A cube with side length L is placed with one corner at the origin and its edges aligned along the x, y, and z axes.



a) (5 points) Your classmate claims that the electric flux passing through each side of the cube must equal zero since it is immersed in a uniform electric field. Are they correct? Why or why not?
b) (5 points) If I were to double the strength of the electric field and keep everything else the same, would the net electric flux passing through the entire cube increase, decrease, or stay the same? Why?
c) (5 points) If I were to shrink the volume of the cube by half and keep everything else the same, would the net electric flux passing through the entire cube increase, decrease, or stay the same? Why?
d) (5 points) If I were to rotate the cube 90 degrees around the z-axis, would the net electric flux passing through the entire cube increase, decrease, or stay the same? Why?